

| Senior Full Stack Software Engineer |



Senior Full Stack Software Engineer specializing in **.NET** and **Angular**, experienced in building **scalable, distributed, and cloud-native applications**. Experienced with **Microsoft Azure, AI services, microservices, CI/CD**, and modern software architecture, with a focus on **Clean Architecture**, maintainable code, and automated testing. I place strong emphasis on understanding the **business domain** and the problem being solved, not just the technical implementation. I actively use **AI-assisted development** and modern AI services to improve engineering productivity and explore practical solutions.

AREAS OF EXPERTISE

Languages: C#, TypeScript, JavaScript

Backend: .NET, ASP.NET Core, Entity Framework Core

Frontend: Angular, HTML, CSS/SCSS

Databases & Caching: Microsoft SQL Server, Oracle, Redis

Cloud & DevOps: Microsoft Azure, Azure DevOps, Bicep, Azure Functions, Azure Container Apps, Azure Key Vault, Azure SQL, Jenkins, GitHub Actions, CI/CD, Cloud-Native Development

Architecture & Engineering: Clean Architecture, Domain-Driven Design (DDD), CQRS, SOLID principles, Microservices, Distributed Systems, Messaging & Event-Driven Architecture

Testing: Unit, Integration, End-to-End (E2E), and Load Testing

AI: Generative AI, AI-Assisted Development, GitHub Copilot, Claude, LLM Integration, RAG, Azure AI Services

PORTFOLIO

Full Stack Finance Application (.NET, Angular, Azure, AI)

Built a **full-stack financial application** with secure authentication, transaction management, CSV import, spending analytics, and AI-powered transaction processing. Designed the Angular frontend from scratch and developed the .NET backend using **CQRS, Domain-Driven Design (DDD), Clean Architecture, and SOLID principles**.

Implemented a **cloud-native architecture** with microservices, messaging, background processing, caching, and REST APIs. Deployed using **Azure Container Apps and Azure SQL**, with **Azure Key Vault** for secrets management, **Bicep** for infrastructure as code, and **Azure AI Services** for AI functionality. Built **CI/CD pipelines with GitHub Actions** and automated testing, demonstrating practical experience with **Azure, distributed systems, DevOps, and AI integration**.

Personal Developer Portfolio (.NET, Angular, Azure AI)

Built and deployed a **personal developer portfolio** with an **Angular frontend** and **.NET backend running on Azure Functions**. Implemented an **AI-powered chat** using **Azure AI Search and Microsoft Foundry**, with **RAG, embeddings, and vector search** to answer questions about my experience, skills, and projects based on portfolio content.

Explore my projects: <https://szilardferencz.dev/#blog>

PROFESSIONAL EXPERIENCE

Senior Software Engineer | Capgemini Hungary Kft / Morgan Stanley | Jan 2026 – Present

- Work as part of the firm's central **.NET Platform Team**, developing and supporting **shared developer infrastructure, libraries** and **internal tooling** used by multiple development teams across the firm, spanning **.NET Framework 4.8** and **modern .NET** applications.
- Build and maintain automated delivery workflows using internal deployment platforms, **TeamCity, Jenkins** and **GitHub Actions**, improving release reliability and streamlining **CI/CD** processes.
- Develop and support internal applications used across the organisation, delivering solutions that improve **business workflows and engineering efficiency**.
- Contribute to the development of **FDC3 libraries for .NET applications**, enabling interoperability and integration between **financial desktop applications**.
- Collaborate with .NET development teams through **technical guidance, code reviews** and **engineering support**, promoting maintainable code and **software engineering best practices**.

Key Accomplishments:

- Redesigned the **Shell library** to run embedded applications in **separate processes**, improving application isolation and the reliability of the host environment.
- Supported the **firm-wide migration** of .NET applications from a legacy file-sharing based system to an **Artifactory-based solution**, modernising application distribution and **dependency management**.
- Delivered an internal training session for **20 software engineers** on **Generative AI and GitHub Copilot**, demonstrating practical applications of **AI-assisted development** and helping teams adopt modern AI development practices.

Full Stack Software Engineer | Robert Bosch Kft. | Jan 2021 – Dec 2025

- Developed and maintained scalable **.NET** backend and **Angular** frontend applications for factory production systems, supporting automation, monitoring, and operational efficiency through **REST API integrations** with external services.
- Designed and implemented **microservices-based architectures** with **Oracle** and **RabbitMQ**, ensuring **high availability** and fault-tolerant production environments.
- Implemented comprehensive **unit, integration, and end-to-end (E2E) testing** using xUnit, NUnit, Cypress, Postman, and Locust to guarantee software reliability, maintainability, and quality.
- Contributed to **system design and architecture** discussions, shaping application structure, feature development, and technical strategy in collaboration with cross-functional teams.
- Delivered **responsive Angular interfaces**, translating Figma designs into maintainable and user-friendly UI.
- Mentored developers and introduced modern **Angular features (Signals, NgRx)**, enhancing team productivity and code maintainability.

Key Accomplishments:

- Refactored legacy code using **Domain-Driven Design principles** and **Entity Framework**, improving maintainability, scalability, and performance.
- Developed a **foundational engineering framework** with configurable steps, enabling flexible, scalable application workflows.
- Optimized **CI/CD pipelines** using **Azure DevOps, Docker, and Kubernetes**, achieving a 40% faster deployment runtime.
- Improved **project and system documentation** using **GitHub Copilot** to streamline writing, maintain consistency, and reduce manual effort.

PLC Programmer | Robert Bosch Kft. | Feb 2018 – Jan 2021

- Developed and maintained **control software for industrial machines** in factory production environments.
- Commissioned machines and supported trial production, ensuring operational efficiency and minimal downtime.
- Collaborated with customers to gather requirements and ensure satisfaction.
- Managed small-scale projects, overseeing timelines, deliverables, and smooth integration with production systems.

Key Accomplishments:

- Contributed to the **implementation and commissioning of a new manufacturing line**, enabling smooth integration with existing systems and operational efficiency.

PLC Programmer | Agrometal-Food-Tech Kft. | Jun 2015 – Feb 2018

- Implemented machine control software from scratch and created **circuit diagrams** for machine systems.
- Participated in commissioning to ensure proper functionality and performance optimization.
- Provided after-sales support and troubleshooting for production systems.

Design Engineer | Baldwin UV Ltd. | May 2014 - May 2015

- Learned core concepts in **control systems, automation, and programming**.
- Applied **signal processing, electrical machinery, and digital system** knowledge to practical projects.

EDUCATION & CERTIFICATIONS

BSc Electrical Engineering | University of Pecs | 2014

LANGUAGES

Hungarian | native English | fully proficient